Somanshu Mohapatra

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SKILLS

Languages: C++, Python, Embedded C, MATLAB (Simulink, Robotics Toolbox).
Robotics: Forward/Inverse Kinematics (DH Parameters), ROS2, Gazebo, Path Planning, Motion Control, PID Control, Sensor Fusion, SLAM.
Electronics: Arduino UNO Q, Raspberry Pi, STM32, ATmega328P, RP2040
Motor Drivers (DRV8825, TB6600), Serial Communication (UART, I2C, SPI, CAN).
CAD & Design: SolidWorks, Fusion 360, AutoCAD, Creo Parametric, FEA, Assembly Design, Rapid Prototyping, 3D Printing (FDM/SLA).
Soft Skills: Effective Organizer, Team Management, Strong Interpersonal Skills, Problem Solving.

INTERNSHIP

AutomateX – PLC based training for Industrial Automation Jun' 25 – Jul' 25
Student Intern | Ladder Logic, LogixPro, TwidoSuite

- Engineered an automated feeder control system in LogixPro using ladder logic, reducing manual labor by 70% and feed wastage by 25%.
- Implemented safety interlocks, emergency-stop recovery, and auto/manual modes to ensure zero-failure restart behavior and reliable operation.

PROJECTS

Dynamic Audio Monitoring and Narration | *Raspberry Pi, Python, AssemblyAI API* Apr' 25 – May' 25

- Developed a Python-based real-time transcription system for the deaf using Raspberry Pi 4B, achieving 95%+ accuracy with <300ms latency.
- Integrated AssemblyAI WebSockets with an I2C SSD1306 OLED display using multi-threading to ensure seamless, non-blocking subtitle updates.

3-DOF Robotic Arm with Real-Time Control | *MATLAB, SolidWorks, Arduino IDE, Bambu Studio* Apr' 24

- Constructed a 3-DOF robotic arm using NEMA 17 steppers and Arduino Mega, utilizing 3D-printed components to optimize payload-to-weight ratio.
- Implemented forward and inverse kinematics via MATLAB-UART communication, reducing trajectory error by 20% during real-time tasks.

Smart Energy Meter | *ESP32, IoT Monitoring, HTML Webserver* Jun' 23

- Designed an IoT-based energy monitoring system using ESP32 and Wi-Fi to track power consumption via a real-time browser interface.
- Integrated ZMPT101B and ZMCT103C sensors to measure voltage and current with ±3% accuracy visualized via custom HTML dashboard.

CERTIFICATES

Data Base Management System | *NPTEL* Aug' 25
AutomateX: Mastering PLCs for Industrial Automation | *LPU* Mar' 25
Fusioncraft | *Resolute Lab* Mar' 24

ACHIEVEMENTS

METHOD FOR ENHANCING FOOD ACCESSIBILITY, MONITORING FRESHNESS, AND REDUCING ENERGY LOSS IN CYLINDRICAL REFRIGERATION UNITS Sep' 25
Somanshu Mohapatra, Dr. Sumit Shoor
Application No.: 202511085342 A

EDUCATION

Lovely Professional University Phagwara, Punjab
Bachelor of Technology Aug' 23 – Present
Robotics and Automation Engineering; CGPA: 8.38

Shri S.N. Siddeshwar Public School Gurugram, Haryana
Intermediate, PCMB; Percentage: 85.6% Mar' 21 – May' 23

Ryan International School Gurugram, Haryana
Matriculation, Percentage: 81.6% Mar' 20 – May' 21